# Proposed Methodology for Deepfake Audio Detection

Teacher–Student Learning Framework with Knowledge Distillation

### 1 ASVspoof 2019 LA Dataset

- Bona fide and spoof speech samples
- Training and development with known attacks
- Evaluation with mostly unseen attacks

### 2 Audio Preprocessing

- Mono conversion
- 16 kHz resampling
- Waveform normalization
- Variable-length handling
- Padding and masking

### 3 Training Augmentation

- RawBoost ① + ②
- Linear and non-linear distortion
- Impulsive signal-dependent noise
- Codec augmentation

### 4 Teacher Model

- Wav2Vec2-XLSR front-end
- AASIST back-end
- P2SGrad / OC-Softmax loss

### 5 Teacher Soft Predictions

- Class confidence scores
- Soft labels for student training

### 6 Student Model

- Frozen Wav2Vec2-XLSR front-end
- AASIST-L lightweight back-end

### 7 Knowledge Distillation Training

- Hard label loss
- Knowledge distillation loss
- Teacher knowledge transfer

### 8 Final Prediction

- Bona fide
- Spoof

### 9 Evaluation

- EER and min-tDCF
- Accuracy, precision, recall
- F1-score and confusion matrix

Output: Bona fide / Spoof classification evaluated using EER, min-tDCF, accuracy, precision, recall, F1-score, and confusion matrix.